

Evaluating large language models for selection of statistical test for research: A pilot study

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Abstract

Background: In contemporary research, selecting the appropriate statistical test is a critical and often challenging step. The emergence of large language models (LLMs) has offered a promising avenue for automating this process, potentially enhancing the efficiency and accuracy of statistical test selection.

Aim: This study aimed to assess the capability of freely available LLMs – OpenAI's ChatGPT3.5, Google Bard, Microsoft Bing Chat, and Perplexity in recommending suitable statistical tests for research, comparing their recommendations with those made by human experts.

Materials and Methods: A total of 27 case vignettes were prepared for common research models with a question asking suitable statistical tests. The cases were formulated from previously published literature and reviewed by a human expert for their accuracy of information. The LLMs were asked the question with the case vignettes and the process was repeated with paraphrased cases. The concordance (if exactly matching the answer key) and acceptance (when not exactly matching with answer key, but can be considered suitable) were evaluated between LLM's recommendations and those of human experts.

Results: Among the 27 case vignettes, ChatGPT3.5-suggested statistical test had 85.19% concordance and 100% acceptance; Bard experiment had 77.78% concordance and 96.3% acceptance; Microsoft Bing Chat had 96.3% concordance and 100% acceptance; and Perplexity had 85.19% concordance and 100% acceptance. The intra-class correction coefficient of average measure among the responses of LLMs was 0.728 (95% confidence interval [CI]: 0.51–0.86), $P < 0.0001$. The test–retest reliability of ChatGPT was $r = 0.71$ (95% CI: 0.44–0.86), $P < 0.0001$, Bard was $r = -0.22$ (95% CI: -0.56 – 0.18), $P = 0.26$, Bing was $r = -0.06$ (95% CI: -0.44 – 0.33), $P = 0.73$, and Perplexity was $r = 0.52$ (95% CI: 0.16–0.75), $P = 0.0059$.

Conclusion: The LLMs, namely, ChatGPT, Google Bard, Microsoft Bing, and Perplexity all showed $>75\%$ concordance in suggesting statistical tests for research case vignettes with all having acceptance of $>95\%$. The LLMs had a moderate level of agreement among them. While not a complete replacement for human expertise, these models can serve as effective decision support systems, especially in scenarios where rapid test selection is essential.

Keywords: Accuracy, artificial intelligence, data analysis, large language models, research methodology, statistical test selection

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INTRODUCTION

In contemporary research, the selection of the appropriate statistical test plays a pivotal role in the rigor and validity of scientific investigations. Researchers routinely face the challenge of identifying the most suitable statistical tests to analyze their data accurately. This process demands a deep understanding of both the data characteristics and the research objectives.^[1] The complexity of this task, compounded by the diverse array of statistical tests available, can lead to errors, suboptimal analysis, and inefficiencies in the research process.^[2]

Advancement in generative artificial intelligence has presented many large language models (LLMs) free to any user. Their simple chatbots made them easily accessed by the researchers. These models serve as transformative decision-support tools that streamline and fortify the research process in several crucial ways.^[3] They bring efficiency and speed to the rapid analysis of complex research scenarios and receive precise recommendations within seconds. This not only saves valuable time but also allows researchers to focus on other critical aspects of their work.^[4,5]

Researchers require robust, efficient, and accurate tools to aid them in the selection of statistical tests that align with their data and research goals. The advent of LLMs offers a promising avenue for automating and streamlining this critical aspect of research methodology.^[6] However, the accuracy of the models in the suggestion of statistical tests was not tested by any previous study.

In this context, the primary aim of this study was to evaluate the capacity of LLMs to suggest suitable statistical tests for research scenarios. By comparing the model's recommendations with those made by human experts, we aimed to determine the accuracy and reliability of LLMs in this critical research task.

MATERIALS AND METHODS

Type and settings

This research is an empirical study aimed at evaluating the capability of LLMs in recommending suitable statistical tests for research. The study was conducted in a controlled experimental setting using online LLMs—OpenAI's ChatGPT3.5, Google Bard Experiment, Microsoft Bing Chat (Creative), and Perplexity. The study was conducted on September 1, 2023–September 15, 2023.

Case vignette preparation

A total of 27 case vignettes were prepared to represent common research models and scenarios that would require

statistical testing. Each vignette was designed with a specific research question for which an appropriate statistical test needed to be determined. The content of these case vignettes was formulated based on previously published literature, ensuring that the information presented in each vignette accurately reflected real-world research scenarios.^[7,8] Before conducting the experiment, each case vignette was reviewed and the content was validated by an expert statistician with >10 years of experience in handling research data to ensure the accuracy of the information and the appropriateness of the statistical test questions.

Large language model selection

Four freely available LLMs were selected for this study: OpenAI's ChatGPT3.5, Google Bard, Microsoft Bing Chat, and Perplexity. These four are freely available LLMs that can be used through company-provided chatbots on their websites and are being used in medical research. The researchers obtained access to these LLMs through their respective online platforms (not application programming interface (API)) to interact with the models.

Accuracy test

Each LLM was presented with the 27 prepared case vignettes, one at a time. For each vignette, the LLM was asked to recommend a suitable statistical test to address the research question presented in the vignette. The recommendations provided by each LLM were recorded and stored for analysis. The responses were compared with the answer keys prepared with the case vignettes with concordance and acceptance.

Concordance refers to the degree of agreement (exact match) between the recommendations made by LLMs with the predefined answer of the vignette. Acceptance is the level of alignment (when not exactly matching with answer key, but can be considered suitable) between the recommendations provided by LLMs and those made by human experts.

Reliability test

The case vignettes were paraphrased while keeping the conceptual equivalence between the initial cases and paraphrased cases. These LLMs were asked to recommend a suitable statistical test to address the research question presented in the paraphrased case vignettes. To determine the reliability of LLM recommendations, the concordance between the recommendations made on the initial set and the paraphrased set (after a gap of 7 days) for each LLM was compared.

Statistical analysis

Descriptive statistics, such as frequencies and percentages, were used to summarize the results. Responses were

categorized into concordance and acceptance and were compared by Chi-square test. The agreement among LLMs was tested by the intra-class correlation coefficient and test–retest agreement was tested by Spearman’s correlation coefficient. We used GraphPad Prism 9.5.0 (GraphPad Software, Boston, MA, USA) for conducting statistical tests. $P < 0.05$ was considered statistically significant.

Ethics

The case vignettes used in this study were available in the public domain. We paraphrased the vignettes for this study. Hence, no real or sensitive data were collected or used for this study. No human research participants were recruited for this study. Hence, this study is exempted from ethics committee review according to Indian Council of Medical Research (ICMR) guidelines.

RESULTS

In our study involving 27 case vignettes, we assessed the performance of four different LLMs in suggesting statistical tests for various scenarios. ChatGPT3.5 exhibited a performance with an 85.19% concordance rate and achieved a remarkable 100% acceptance rate. Google Bard experiment achieved a 77.78% concordance rate and acceptance rate of 96.3%. Microsoft Bing Chat displayed a 96.3% concordance rate and a 100% acceptance rate. The Perplexity demonstrated an 85.19% concordance rate and a 100% acceptance rate. The result is shown in Figure 1 ($\chi^2 = 0.38$, $P = 0.53$). There was no significant difference among the performance of the LLMs.

The calculated ICC was 0.728, with a 95% CI 0.51–0.86, indicating a moderate degree agreement among the LLMs in suggesting statistical tests. This level of agreement was statistically significant, with a $P < 0.0001$.

The responses in the test and retest are shown in Table 1. The test–retest reliability of ChatGPT was $r = 0.71$ (95% CI: 0.44–0.86), $P < 0.0001$, Bard was $r = -0.22$ (95% CI: -0.56–0.18), $P = 0.26$, Bing was $r = -0.06$ (95% CI: -0.44–0.33), $P = 0.73$, and Perplexity was $r = 0.52$ (95% CI: 0.16–0.75), $P = 0.0059$. Hence, the test–retest reliability was high in ChatGPT and Perplexity.

DISCUSSION

In our study, assessing the performance of four different LLMs in suggesting statistical tests for various scenarios, we found that ChatGPT3.5, Google Bard Experiment, Microsoft Bing Chat, and Perplexity all provided similar levels of performance. These LLMs offered reliable suggestions for statistical tests, with no significant

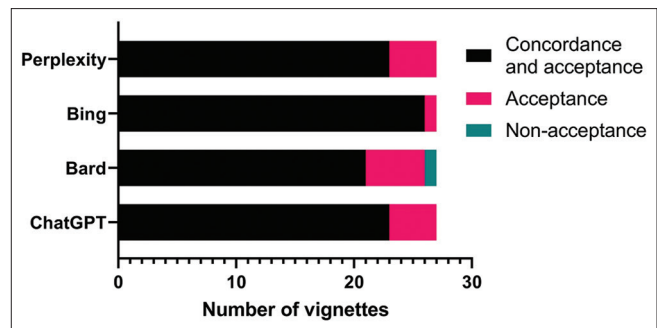


Figure 1: Visual representation of the performance of four large language models in answering statistical case vignettes

differences among them. Users can choose any of these models based on their preferences and needs. This study shed light on the potential role of freely available LLMs in selecting appropriate statistical tests in research. Researchers from developing countries would be benefitted from those LLMs in getting suggestions for their research models.^[9,10]

In test–retest, the ChatGPT and Perplexity consistently offered similar recommendations. On the other hand, Bard and Bing displayed fewer deviations in their recommendations, resulting in less consistent suggestions across the two testing phases. The reason behind this finding can likely be attributed to the inherent variability in how different LLMs process and interpret information. This consistency could vary according to the architecture or training methodology.^[11,12] Furthermore, factors such as training data, algorithms, or model complexity may contribute to the varying levels of consensus. However, this finding indicates that changing the words in a vignette although the conceptual equivalence is there, the response of the LLMs may deviate.

The response styles of the four LLMs are different as these are developed by four companies and their architecture is different or tuned for specific purposes. For example, we created the following vignette and associated question – “To capture the photomicrographs, a low-cost smartphone adaptor was created. Histopathological slides from 20 patients were collected using the adaptor and forwarded to two pathologists who used the photomicrograph to diagnose the condition. A table was created with the following categories: both observers diagnosed, the first observer diagnosed and the second observer did not diagnose, the first observer did not diagnose and the second observer diagnosed, and both observers did not diagnose. Suggest single best statistical test for agreement.” When LLMs were asked to solve the problem, the responses were different and part of the replies in part from ChatGPT is

Table 1: Concordance and acceptance of answers provided by large language models in test and retest

Category	ChatGPT3.5, n (%)	Bard experiment, n (%)	Bing creative, n (%)	Perplexity, n (%)	P
Test (n=27)					
Concordance and acceptance	23 (85.19)	21 (77.78)	26 (96.3)	23 (85.19)	0.53
Acceptance	4 (14.81)	5 (18.52)	1 (3.7)	4 (14.81)	
Nonacceptance	0	1 (3.7)	0	0	
Retest (n=27)					
Concordance and acceptance	23 (85.19)	23 (85.19)	24 (88.89)	24 (88.89)	0.95
Acceptance	4 (14.81)	4 (14.81)	3 (11.11)	3 (11.11)	
Nonacceptance	0	0	0	0	

P value is of Chi-square test

shown in Figure 2, Bard is shown in Figure 3, Bing is shown in Figure 4, and Perplexity is shown in Figure 5.

The novelty of this study lies in its comprehensive assessment of four different LLMs in the context of suggesting statistical tests. The study implications are significant, as it provides users with the knowledge that these LLMs exhibit comparable levels of performance in suggesting statistical tests. This insight can empower researchers, educators, and professionals to select the most suitable LLM for their specific needs and preferences of their statistical analysis processes. In addition, this study underscores the potential of LLMs in assisting with complex tasks in the domain of data analysis and statistical decision-making, paving the way for their broader utilization in research and practical applications.

One significant limitation of this study is the usage of only publicly available LLMs, which may not fully represent the broader landscape of AI models. The evaluation of LLM performance was based on a relatively small set of 27 case vignettes, which may not capture the full spectrum of research scenarios. Furthermore, the study assessed LLM recommendations over a brief 7-day interval, and the temporal stability of LLM performance was not explored extensively. Finally, the study findings may not generalize to more complex or specialized research domains, as it focused on common research models. These limitations underline the need for caution in relying solely on LLMs for critical research decisions and highlight the importance of ongoing evaluation and transparency in their use.

CONCLUSION

We found that ChatGPT3.5, Google Bard Experiment, Microsoft Bing Chat, and Perplexity LLMs exhibited comparable levels of performance, with high acceptance rates and concordance rates, indicating their ability to provide reliable recommendations for statistical tests. The study's findings also highlighted a moderate degree of agreement among these LLMs, further underlining their consistency in this task. ChatGPT3.5 and Perplexity

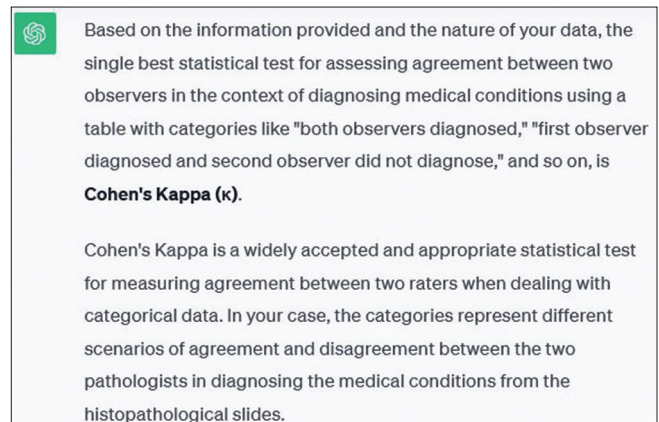


Figure 2: Part of response by ChatGPT captured from <https://chat.openai.com> during a conversation

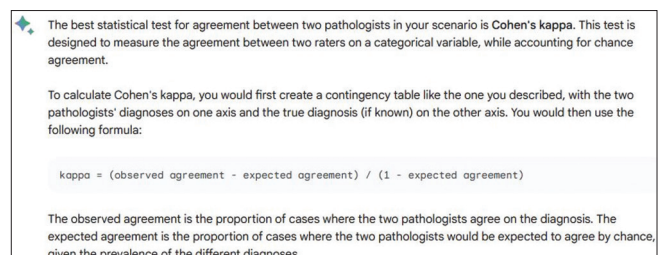


Figure 3: Part of response by Google Bard captured from <https://bard.google.com> during a conversation

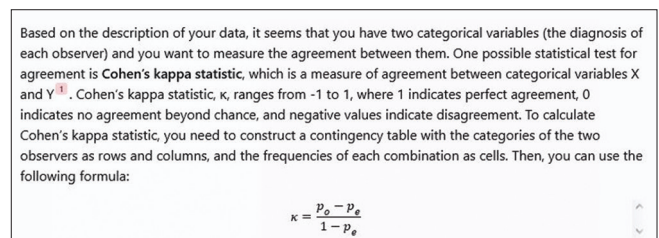


Figure 4: Part of response by Microsoft Bing Chat (GPT4) captured from <https://www.bing.com> during a conversation

demonstrated high test–retest reliability, whereas Bard and Bing exhibited lower levels of consistency in their recommendations. The absence of significant differences in their overall performance implies that users can confidently choose any of these LLMs based on their preferences and requirements. This study contributes to

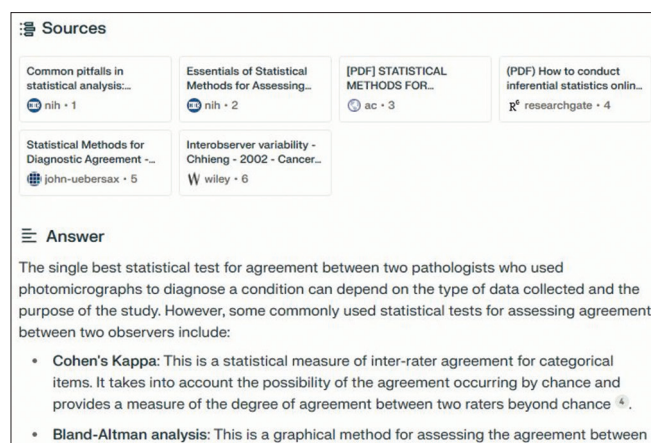


Figure 5: Part of response by Perplexity.ai captured from <https://www.perplexity.ai> during a conversation

our understanding of the capabilities of LLMs in aiding statistical decision-making and underscores their potential as valuable tools in data analysis and research.

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Conflicts of interest

There are no conflicts of interest.

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